

HCI and AI in Industry: Current and Future

Jofish Kaye
Wells Fargo
acm@jofish.com

Victoria Bellotti
Bloomberg
vbellotti@bloomberg.net

Jaime Teevan
Microsoft
teevan@microsoft.com

Lauren Wilcox
EBay
l.g.wilcox@gmail.com

ABSTRACT

In this panel, composed of leading members of the field from industry, we consider impacts of AI technologies, and how HCI impacts developments in AI. Our focus on industry recognizes the many ways in which AIs underlie familiar and novel product and service experiences, and how HCI practitioners ensure that human concerns such as privacy, safety, diversity, equity and inclusion are prioritized. We articulate key areas of AI impact, and detail problem areas that the discipline of HCI can assist with. In discussion between panelists and the audience, we will focus mainly on those areas: namely, the invisible “jagged edge” of AI-supported human-computer interaction; the impact of training and user input datasets in driving outputs; and concerns about AI applied to personal data. We further explore the implications of all three of these for HCI as a field, and the vital impact of HCI on AI.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); • **Computing methodologies** → Artificial intelligence.

KEYWORDS

HCI, AI, industry, LLMs, practice

ACM Reference Format:

Jofish Kaye, Jaime Teevan, Victoria Bellotti, and Lauren Wilcox. 2024. HCI and AI in Industry: Current and Future. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3643974>

1 INTRODUCTION

Recent AI developments such generative AI, large language models (LLMs), and more, have major implications for the discipline of human-computer interaction. These impact both how we professionals develop products and solutions and how people experience them on a day-to-day basis. There are multiple stakeholder interests to consider including original content creators; the people from whom AI learns; developers and sellers of AI-based products and

services; end users who actively engage with AI for specific purposes (of which we may or may not approve); and those whose lives are impacted by AI, even if they have not actively or voluntarily engaged with it.

In this panel, senior industry researchers involved in both HCI and AI will address the use and impacts of AI-based technologies from an HCI perspective and discuss directions for future developments in both fields. To set the scene, we will discuss applications of AI in our community’s practices of both qualitative and quantitative research and in design and implementation of technology solutions. We also offer some examples of AI applications where panelists have relevant experience and where HCI as a discipline must play a significant role if the benefits are to be fully realized and the risks minimized. The examples are intended to provide context for a thought-provoking debate. Having set the scene, we will focus on three interesting problem areas of great or even urgent interest to the research and industrial HCI communities. We will use these to seed a lively discussion including the perspectives of both panelists and audience members.

2 HCI AND AI IN INDUSTRY: CURRENT PRACTICES AND LOOKING TO THE FUTURE

In this panel, we propose to present viewpoints from industry practitioners who have experience of active involvement in the use and development of a wide variety of AI tools. This will be a highly interactive, dynamic panel with provocations and questions posed by the Chair, the panelists and audience, starting from but not limited to the themes we have introduced. We intend to have only the briefest of introduction statements from the panelists, and a great deal of audience involvement. We would expect the panel to attract a diverse and enthusiastic audience: people interested in the use of AI to support their own practices, people currently working in AI with an interest in the role of HCI, and, perhaps, people who disagree with the use of AI in the context of HCI, all of which suggest it will be a lively, provocative, and above all interesting panel.

We intend to enthusiastically engage with our audience, before and during the panel. We look forward to understanding the exact infrastructure provided by the conference for engagement, but we will leverage social media tools like Discord, Mastodon, BlueSky, and Facebook to encourage discussion and generation of questions, as well as using a system like sli.do for real-time participant involvement.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3643974>

3 EXAMPLES OF HCI-RELATED AI APPLICATIONS

3.1 Examples of AI Applications for HCI, UX, and Strategic New Product Development

We begin our collection of examples to contextualize this panel discussion with some that pertain to our own practice, and the questions they raise. How do they change our jobs and the ways work is performed? Should entities share ways to use AI in the practice of HCI, as the team at Microsoft has done with the HAX toolkit [13]? Or should they offer the ability to create HCI-specific AI models to researchers (and anyone else) as a service for a fee as Open AI has done with its GPT Builder [15]? Should they use AI to compete in building better experiences, meaning that the few companies that control truly powerful AI models will increase their dominance? In this section, we discuss four domains where research suggests thought-provoking opportunities for the practice of HCI, UX, and strategic new product development in both academia and industry.

3.1.1 UX Research Data Gathering and Market Research. AgentGPT [2] can be used to guide the development of a research study, from the process to the actual interview guide. Since planning research is an involved process, users may have to break the plan prompts down into stages. For a junior researcher, still learning the ropes, this can reduce the effort of devising a plan and the risk of missing important steps in the process. tldv [21] is a GPT-enhanced tool that can take live notes and summarize interview insights in qualitative research. Such tools can drastically reduce certain tedious steps in qualitative research, condensing large amounts of text down to the valuable nuggets that drive insights. With the time saved, researchers can afford to include a larger sample of participants, perhaps extending to normally overlooked minorities such as demographic outliers or people with disabilities.

Similarly, Brand et al [5] showed that ChatGPT 3.5 can be leveraged to reduce effort in market research, in that it “responds to sets of survey questions in ways that are consistent with economic theory and well-documented patterns of consumer behavior, including downward-sloping demand curves and state dependence”. So, when investigating market readiness for a product concept, LLMs may become useful in validating or falsifying assumptions about the target market.

3.1.2 UX Research Analysis. Bano et al [3] ran an experiment to compare ChatGPT 3.5 and 4 with humans qualitatively classifying Amazon reviews and found that the LLMs aligned with humans in between a quarter and a third of cases and showed promise as tools to assist human classifiers. Classification of qualitative data is a significant aspect of UX research so the implications here for AI’s potential are fairly obvious. De Paoli [7] ran an experiment to use GPT 3.5-Turbo “to emulate some aspects of an inductive Thematic Analysis.” and showed how a “model can infer at least partially some of the main themes” generated by humans, concluding that “some elements of LLM data manipulation can to an extent be of support for qualitative research.”

3.1.3 UX Design and Evaluation. There are many examples now of commercial website builders (e.g., [18]) which can drastically reduce

the barriers to and cost of developing websites for those who don’t need anything particularly custom and are willing to have a site that depends on the website builder’s platform remaining available. ChatGPT can also automate many aspects of the design process [10], even allowing users to ask it to build custom GPTs to help with the UX design process like UX Design GPT [16]. Similar processes have been offered for evaluation: Synthetic Users [20] is a service that, according to its own website, “enables companies to conduct user research without the typical headaches associated with recruiting participants. The tool offers synthetic users, which are AI-driven virtual participants, to test products, ideas, and concepts.” WebScore AI [23] used machine learning to evaluate 125 parameters that expert designers identified as being used by visitors to evaluate the quality of websites. It has built an AI that can be used for automatic evaluation. AccessibBe.ai [1] has an AI trained on past user behavior, that is used for “complex accessibility adjustments such as screen-reader optimization and keyboard navigation.”

3.2 AI Applications in Industry

Moving from our own professional territory outwards to industry at large, your favorite LLM will happily give you a long list of areas where AI is having the biggest impacts. But, for this panel proposal, we will describe a few that the panelists can speak directly to and that have been gaining attention recently. In this section we list five domains that present opportunities that are key areas for a human-centered approach to AI.

3.2.1 Knowledge Worker Productivity. Some of the greatest strengths of the LLM-based applications that have recently been making headlines are searching, summarization, and composition. Capabilities that can reduce the effort involved in a significant proportion of the work of many knowledge workers. To date, the implication is that the benefits are significant, for example, one (as-yet-to-be-peer-reviewed) study of Boston Consulting Group consultants showed consultants using AI were significantly more productive (they completed an average of 12% more tasks, and finished tasks 25% more quickly), and at higher quality than controls [8].

Microsoft’s Copilot team, led by panelist Jaime Teevan, recently conducted 30 studies [6] to assess their LLM-based assistant, Copilot’s performance in enhancing productivity on various “tasks on the AI-friendly side of the ‘jagged technological frontier’” (more on this term later). They found that, *for AI-compatible tasks*, Copilot tends to confer a “meaningful increase in speed of execution [~25-75% reduction in time taken] without a significant decrease in quality” and that its use increased users’ willingness to pay for LLM-tool support. The cited report on these studies does, however, warn that the tasks tested were cherry-picked and that the same advantages with Copilot might not extend to other types of tasks.

3.2.2 Customer Service. Many companies have been using natural language processing (NLP) or LLM technologies to assist or replace customer service agents for quite some time now [12]. And the trend continues today [9]. In more than one panelist’s company, chatbots handle a significant minority of customer service calls unaided. AI is also used to suggest or support search for relevant responses to others that can be referenced by a human agent.

Notwithstanding the debate about job replacement, from an HCI standpoint, we should be considering how it is possible to greatly enhance the success of support and overall customer experience of a product or service as well as improve the job of the customer support agent with these technologies. As we all know, getting better, faster tech support can greatly enhance our experience with and ability to make the best use of the proliferating technologies that we rely on. What more can be done to support technology users with the power of AI?

3.2.3 Entertainment (Recommenders). AI can play many different roles in the entertainment sphere, many of which have recently inspired controversy. Particularly newsworthy have been ways in which AI can be used to generate scripts, music, visual effects, and animations. These innovations, however, are probably not the most relevant to the HCI profession and our discussion. But another application is predicting what content individuals might be interested in. Panelist Bellotti worked on the UX of Netflix's ranking algorithm (yes, a ranking algorithm really does have a UX) and found that the vast majority of customers she spoke to were not too concerned about the profiling. Indeed, the more the better, when customers are confident it is only being used to improve their experience and is based on information that is not shared outside the service provider.

3.2.4 Self-Driving. Anyone who lives in or has visited San Francisco, Phoenix AZ, or Las Vegas in the past few years, is probably familiar with autonomous vehicles operating within ride hail services, and they are increasingly being seen without a safety driver present in the driver's seat. The experience of riding in a ride-hail AV requires an enormous contribution from HCI practitioners to improve the smoothness of the service and communication with the AV, and to address problems with the ride when they occur (for example, how to handle traffic accidents, passengers becoming ill, or passengers wanting to disembark before the scheduled drop-off).

Many people now own a car that has some level of autonomous AI capability. Autonomy goes from level 1 (driver assistance) up to level 5 (full automation with no driver intervention required). As of 2023, Tesla, General Motors, Audi, BMW, and Volvo all have some level of self-driving, although none is capable of replacing a human in the driver's seat. All these vehicles have what is called an 'operational design domain' (ODD) for self-driving which can be defined as locations and conditions within which the vehicle is thought to be competent to drive itself. Unfortunately, an ODD has a pretty "jagged edge" that is not easy to discern as we shall discuss below.

3.2.5 Software Development. Since this is an ACM Conference, it would probably be remiss of us to omit the huge application opportunity for text to code generation offered by LLMs. At the time of writing, the capabilities of GPT models are limited (e.g. [17]). However, over time we can expect the models to improve. Meanwhile, it is important to remember that the code-generation capabilities depend on the training data set. Bloomberg, for example, has done research on a highly domain-constrained GPT model [25] which, although much smaller than similar-sized models, performed better in tests on domain-constrained financial tasks. Bloomberg has also developed a Text2BQL (Bloomberg Query Language) tool that is

designed specifically for querying its own financial datasets. By limiting the generalizability of the target domain, much smaller and more efficient models can be used. However, for the latter tool, much depends on appropriate UX design to enable users to create suitable prompts and understand and verify that the code they generate is what they really intend.

4 AI PROBLEM AREAS: THE CORE OF THE PANEL

We have given the above examples to set some context for three problem areas that will provide the core of the debate for this panel, chosen as likely to be of high interest to the HCI audience. Those three areas are: the jagged edges of AI; garbage/gold in/out; and the limits of personalization.

4.1 The Invisible Jagged Edges of AI Competency

As Dell'Acqua and his colleagues demonstrated, although they can greatly increase productivity, once LLMs are applied outside their area of competency, they can detract from user performance [8]. We just don't know what the areas of competence are for a given AI. Dell'Acqua et al talk about a "jagged technological frontier" of AI capabilities, echoing Minsky's term "jagged edge" to describe the strengths and weaknesses of human intelligence [14]. But it is not just its being jagged that is the challenge, it is that the edge is invisible to the user. Most AI applications do not communicate their level of confidence in their own performance and give no hint that they might be hallucinating or otherwise confused. This problem is responsible for stories like the recent entertaining news item in which ChatGPT hallucinated case precedents that were used in a lawsuit [24]. Another excellent example is the baby-making problem posted on Twitter/X [22]. For more examples of failures, categorized according to type, see [4].

From our examples above, text-to-code applications and user-activated self-driving mode also have serious invisible-jagged-edge challenges. In user-activated self-driving mode, the consequences are life threatening, since drivers lose situation awareness as they are potentially lulled into a false sense of system competence and allow themselves to be distracted while the AI is in the driver's seat, so to speak. Most users have little to no understanding of how the AV does what it does and are therefore poorly qualified to understand when they are outside of the operational design domain for that vehicle (which may vary even for the same location if lighting, weather and/or road conditions change). There is much to be done here by HCI professionals to address this pervasive problem.

4.2 Garbage/Gold in, Garbage/Gold Out

The matter of AI often being biased is well understood by the HCI community [19]. It's just one symptom of the fact that the outputs that AIs usually generate rely on the information they have been trained on. Along with bias, we have the risk of spitting out information that we don't want in the hands of just anybody, propagation of misinformation, creation of echo chambers of AIs learning from other AI's outputs, copyright infringement, and so on.

For LLMs, one way to combat both the jagged edge and garbage problems at the same time is to provide access to sources to back up assertions that AIs put out. It might also help to provide metadata about those sources if the user cannot be relied upon to be familiar with them: for example, does the information source have a track record of accuracy or honesty or just the opposite? Is the information source knowledgeable about this topic? For self-driving vehicles, solutions to the problem have generally involving carefully constrained sources of training data, such as paid-human tagging of objects that vehicles need to recognize, or using safety drivers to capture data to build a 3D model of the roads and cityscapes that the car can use for more precise positioning than GPS allows.

4.3 When Personalization Gets Too Personal

As mentioned above, profiling may be an acceptable way to improve the experience for the end user. However, when profiling AI is used for advertising and marketing, consumers may be much less comfortable because their profile has now been commoditized and used in the interests of unknown parties including vendors with whom they do not have an existing relationship (see for example, [11]). Given many people do not know what information about them is “out there,” there is a risk that a future AGI (artificial generalized intelligence) operated by a less than benevolent entity is able to pull all that disparate information together and make inferences and predictions. And how much more serious might this be for people who live in a country where human rights are routinely violated by powerful entities such as the state or organized crime?

How can the HCI community help drive solutions, UX designs, and standards and educate consumers about the coming threats to their privacy and public profile in an age where it is possible to create a comprehensive dossier on anyone, and one that might contain grossly inaccurate information?

5 PARTICIPANTS

While we believe the selection of authors from industry instead of academia provides a particular focus and value of this panel to the CHI community, it is not in any way a panel reflecting the opinions of any company, and each of the authors will reflect their own opinions and views, and will not in any way reflect their official policy or positions of any of their employers. We will also repeat this statement at the beginning of the presentation of the panel.

Jofish Kaye, Moderator and Panelist, is a Principal Research Scientist at Wells Fargo in Strategic Design and Innovation, and has previously worked at Anthem.ai, Mozilla, Yahoo and Nokia. His work involves applying AI, behavioral science, and human-centered design to new product and development, and he has an extensive track record of publishing in human-computer interaction (HCI) and computer-supported cooperative work (CSCW). He has served on the ACM Diversity & Inclusion Council, the SIGCHI Executive Board, CRA-Industry, and co-chaired CHI '16. He holds a Ph.D. in Information Science from Cornell University and BS and MS degrees from MIT.

Jaime Teevan, Panelist, is a Chief Scientist and Technical Fellow at Microsoft, where she is responsible for bringing research-backed innovation into the company's core products. She led the creation of Microsoft 365 Copilot, which combines the power of large language

models (LLMs) with the organization's data, by integrating AI into Microsoft's core productivity software. Her research earned her the MIT Technology Review TR35 Young Innovator, and Time Inc. Magazine recently recognized Jaime as one of the 100 Most Influential People in AI for her role in AI development and societal advancement. She holds a Ph.D. in AI from MIT and a B.S. from Yale, and is an Affiliate Professor at the University of Washington.

Victoria Bellotti, Panelist, is a distinguished researcher in the field of Human-Computer Interaction (HCI), specializing in the study of user-centered design and the social implications of technology. With a career spanning several decades, Bellotti has made significant contributions to the understanding of how people interact with and manage their personal information in digital environments. Her expertise includes designing and analyzing systems for collaboration, communication, and personal information management, often focusing on contextual and anticipatory computing. Bellotti has worked at some of the leading technology research organizations, including Xerox PARC, where her work has helped shape the way we think about and interact with technology in our daily lives. Renowned for her pioneering research in activity-centered design, Bellotti's work has not only been influential in academic circles but also has had practical applications in designing user-friendly technology interfaces. Her numerous publications, keynote speeches, and active participation in the HCI community reflect her dedication to advancing the field and mentoring the next generation of HCI professionals.

Lauren Wilcox, Panelist, is Senior Director and Distinguished Applied Scientist in Responsible AI at eBay, where her team conducts research and development at the intersection of multiple disciplines, to shape tools and processes that uphold ethics values and standards and govern the company's approach to AI risk, development, and oversight. Wilcox has held multiple leadership roles in both industry and academia. At Google Research, her team advanced fundamental and applied research that adopted an interdisciplinary lens to engage with, and account for, the social, cultural, economic, and other contextual dimensions that shape the development and deployment of AI systems and their impacts on people. Wilcox is an adjunct associate professor at Georgia Tech's School of Interactive Computing. She holds a Ph.D. in Computer Science from Columbia University.

REFERENCES

- [1] AccessibBe.ai. Retrieved December 10, 2023 from <https://accessibe.com/accessflow/ai-auditing>
- [2] AgentGPT. Retrieved December 10, 2023 from <https://agentgpt.reworkd.ai/>
- [3] Muneera Bano, Didar Zowghi and Jon Whittle. 2023. Exploring Qualitative Research Using LLMs. arXiv:2306.13298. <https://doi.org/10.48550/arXiv.2306.13298>
- [4] Ali Borji. 2023. A Categorical Archive of ChatGPT Failures. arXiv:2302.03494v8 [cs.CL]. <https://github.com/giuven95/chatgpt-failures>
- [5] James Brand, Ayelet Israeli and Donald Ngwe. 2023. Using GPT for Market Research (March 21, 2023). Harvard Business School Marketing Unit Working Paper No. 23-062. <http://dx.doi.org/10.2139/ssrn.4395751>
- [6] Alexia Cambon *et al.* 2023. Early LLM-based Tools for Enterprise Information Workers Likely Provide Meaningful Boosts to Productivity. MSR-TR-2023-43 | December 2023. Retrieved December 12, 2023 from <https://www.microsoft.com/en-us/research/uploads/prod/2023/12/AI-and-Productivity-Report-First-Edition.pdf>
- [7] Stefano De Paoli 2023. Can Large Language Models emulate an inductive Thematic Analysis of semi-structured interviews? An exploration and provocation on the limits of the approach and the model. arXiv:2305.13014v3 [cs.CL]. <https://doi.org/10.48550/arXiv.2305.13014>
- [8] Fabrizio Dell'Acqua Edward McFowland, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Saran Rajendran, Lisa Krayner, François Candelon, and Karim R.

- Lakhani. 2023. Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality (September 15, 2023). Harvard Business School Technology & Operations Mgt. Unit Working Paper No. 24-013. <http://dx.doi.org/10.2139/ssrn.4573321>
- [9] Emma Goldberg. 2023. 'Training My Replacement': Inside a Call Center Worker's Battle With A.I. New York Times. Retrieved December 10, 2023 from <https://www.nytimes.com/2023/07/19/business/call-center-workers-battle-with-ai.html>
- [10] Jeff Humble. 2023. 11 Advanced UX Tasks You Should Start Automating in Chat GPT. Medium. Retrieved December 10, 2023 from <https://medium.com/@jeffreyhumble/11-advanced-ux-tasks-you-should-start-automating-in-chat-gpt-63a6a1b45cc1>
- [11] Nancy J. King and Pernille Wegener Jessen. 2010. Profiling the mobile customer – Privacy concerns when behavioural advertisers target mobile phones – Part I. Computer Law & Security Review. Volume 26, Issue 5, 455-478. <https://doi.org/10.1016/j.clsr.2010.07.001>
- [12] Keith Kirkpatrick. 2017. AI in Contact Centers. Communications of the ACM, 60, 8, 18–19 <https://doi.org/10.1145/3105442>
- [13] Microsoft. HAX Toolkit. Retrieved December 10, 2023 from <https://www.microsoft.com/en-us/haxtoolkit/>
- [14] Marvin Minsky. 1986. Society of Mind. New York: Simon & Schuster. ISBN 0-671-60740-5.
- [15] Open AI. 2023. Creating a GPT. Retrieved December 10, 2023 from <https://help.openai.com/en/articles/8554397-creating-a-gpt>
- [16] Bapusaheb Patil. UX Design GPT. Retrieved December 10, 2023 from <https://bapspatil.gumroad.com/l/ux-design-gpt>
- [17] Jaromir Savelka, Arav Agarwal, Christopher Bogart, Yifan Song, Majd Sakr. 2023. Can Generative Pre-trained Transformers (GPT) Pass Assessments in Higher Education Programming Courses? In Proceedings of ITiCSE 2023 Conference on Innovation and Technology in Computer Science Education V. 117–123. <https://doi.org/10.1145/3587102.3588792>
- [18] Emily Shwake. 2023. 9 best AI website builders for creating professional sites in minutes. WIXBlog. Retrieved December 10, 2023 from <https://www.wix.com/blog/best-ai-website-builder>
- [19] Ramya Srinivasan & Ajay Chander. 2021. Biases in AI systems. Communications of the ACM, 64, 8, 44–49. <https://doi.org/10.1145/3464903>
- [20] Synthetic Users. Retrieved December 10, 2023 from <https://www.syntheticusers.com/>.
- [21] tldv. Retrieved December 10, 2023 from <https://tldv.io/remote-ux-research-tool/>
- [22] Twitter/X Post. 2023. Retrieved December 10, 2023 from <https://twitter.com/KaiaVintr/status/1607453952525881344>
- [23] WebScore AI. Retrieved December 10, 2023 from <https://webscore.ai/>
- [24] Benjamin Weiser. 2023. Here's What Happens When Your Lawyer Uses ChatGPT. New York Times, May 27, 2023. Retrieved December 10, 2023 from <https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>
- [25] Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhajan Kambadur, David Rosenberg, and Gideon Mann. 2023. BloombergGPT: A Large Language Model for Finance. arXiv:2303.17564 <https://doi.org/10.48550/arXiv.2303.17564>